

# TASK7: SMART CITY COMPLAINT MANAGEMENT SYSTEM

## 1. Introduction

Smart cities generate large volumes of citizen complaints related to public services such as water supply, electricity, roads, sanitation, waste management, and street lighting. Efficient complaint management helps city authorities improve service quality and citizen satisfaction.

This project analyzes smart city complaint data using Python and Power BI. The objective is to identify complaint patterns, evaluate department performance, analyze geographic distributions, and support data-driven decision-making through interactive dashboards.

## 2. Business Problem

Smart city administrations receive thousands of complaints every day. Managing these complaints manually creates several challenges:

- High complaint volumes.
- Delayed complaint resolution.
- Uneven department workload.
- Difficulty identifying recurring issues.
- Poor resource allocation.
- Reduced citizen satisfaction.

City administrators require answers to the following questions:

- i. Which complaint categories occur most frequently?
- ii. Which departments are overloaded?
- iii. Which city zones generate the highest complaints?
- iv. Are high-priority complaints resolved quickly?

v. How can dashboards support operational and policy decisions?

### **3. Objectives**

- Analyze smart city complaint data.
- Identify complaint trends and patterns.
- Evaluate department performance.
- Analyze complaint distribution across city zones.
- Monitor resolution efficiency.
- Measure citizen satisfaction.
- Develop interactive Power BI dashboards.
- Provide business recommendations for service improvement.

### **4. Dataset Description**

A synthetic Smart City Complaint dataset containing approximately 1,000,000 records was used.

#### Features

- Complaint\_ID
- Complaint\_Date
- Resolved\_Date
- Category
- Department
- Zone
- Priority
- Status
- Citizen\_Satisfaction
- Resolution\_Delay\_Days

- Complaint\_Description

Each record represents one citizen complaint and its resolution details.

## **5. Tasks to Perform**

### Phase 1: Data Understanding

#### Tasks Performed

- Loaded the dataset.
- Examined sample records.
- Checked data dimensions.
- Verified data types.
- Identified missing values.
- Checked duplicate records.
- Converted date columns.
- Calculated resolution delays.

#### Findings

- Dataset contains approximately 1,000,000 complaints.
- Missing values were handled.
- Duplicate records were removed.
- Date fields were properly formatted.
- Dataset was ready for analysis.

### Phase 2: KPI Identification

#### KPIs Developed

- Total Complaints.

- Resolution Rate.
- Pending Complaints.
- Average Resolution Time.
- Citizen Satisfaction Score.
- Complaint Trends.
- Department Performance.
- Zone-wise Complaint Count.

#### Findings

- KPIs provide an overview of city operational health.
- Decision-makers can monitor complaint handling efficiency.

#### Phase 3: Complaint Investigation

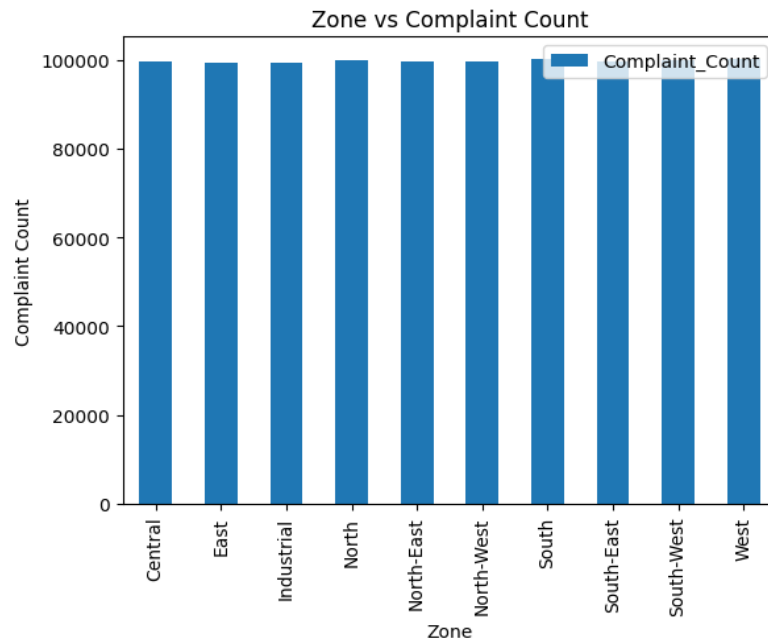
##### Analysis Performed

Relationships between complaints and:

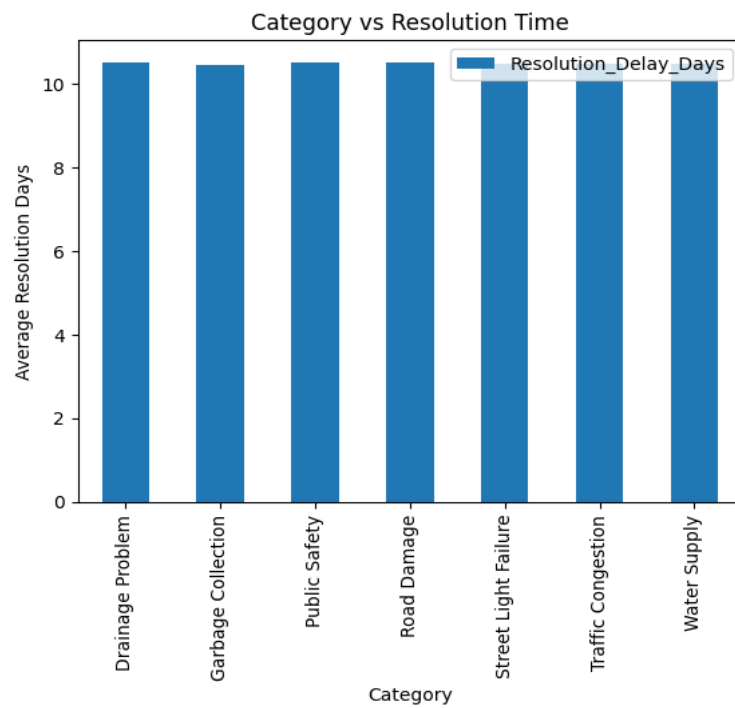
- Complaint Category.
- Department.
- Zone.
- Priority.
- Resolution Delay.
- Citizen Satisfaction.

Visualizations used:

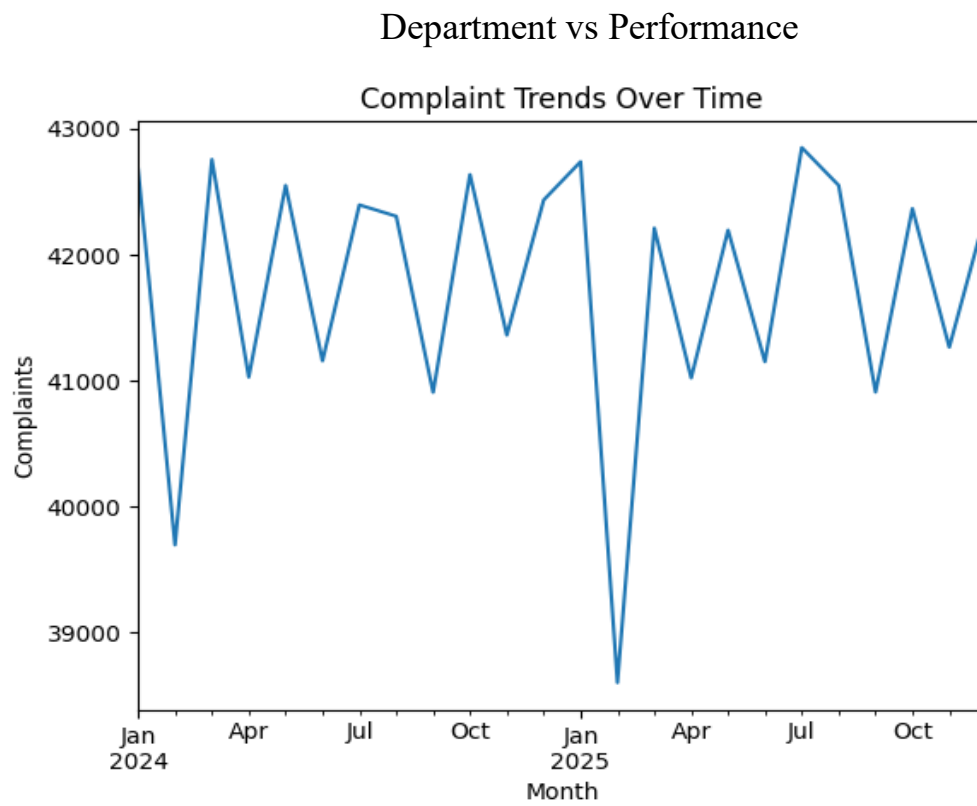
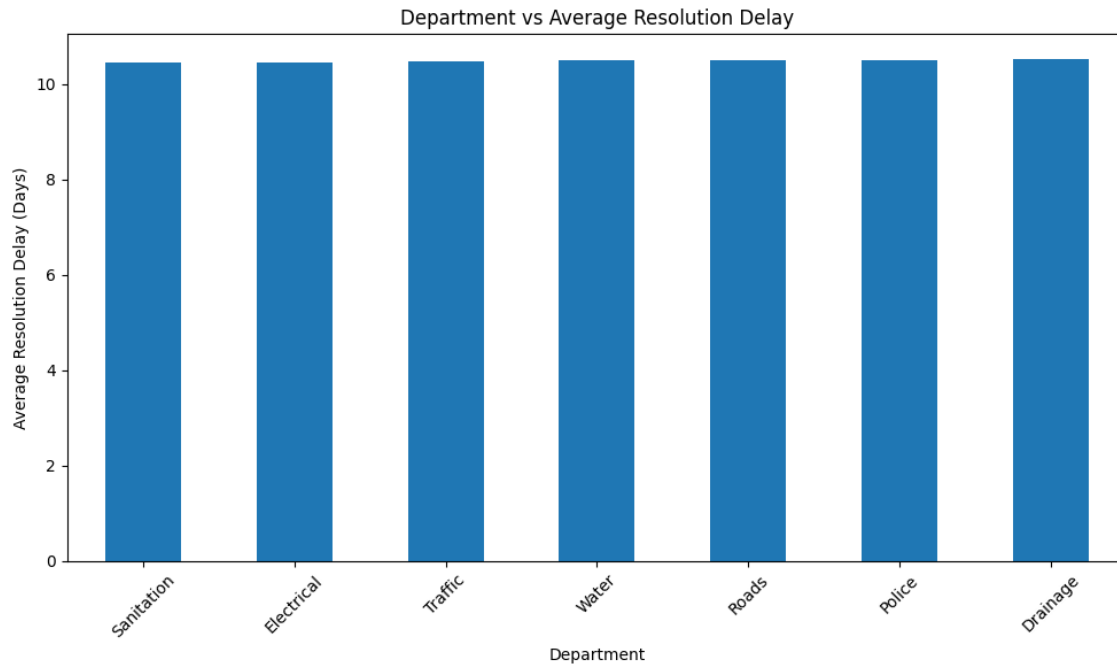
- Bar charts.
- Pie charts.
- Line charts.
- Heatmaps.



Zone vs Complaint Count



Category vs Resolution Time



Complaint Trends Over Time

Findings

- Certain categories generated more complaints.
- Some departments received higher workloads.
- High-priority complaints required immediate attention.
- Citizen satisfaction depended on resolution efficiency.

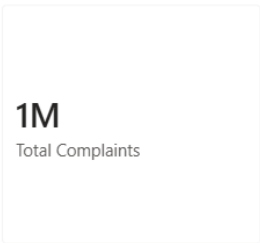
Phase 4: Dashboard Development

Executive Overview Dashboard

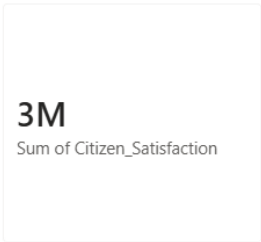
Includes:

- Total complaints.
- Resolution rate.
- Satisfaction score.
- Complaint trends.

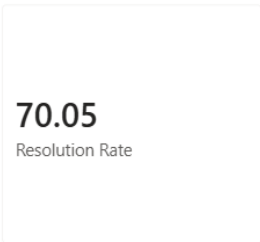
Total complaints



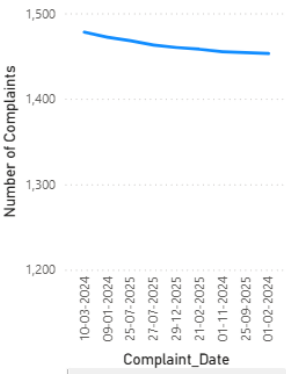
Satisfaction score



Resolution rate



Complaint Trends Over Time

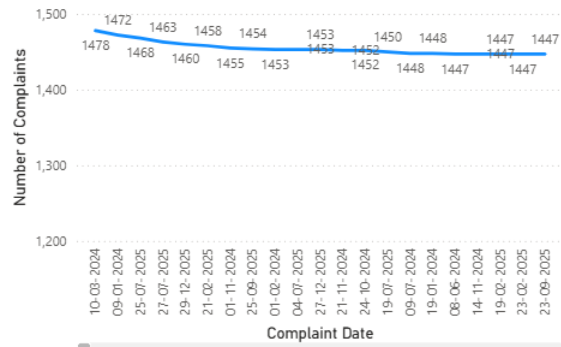
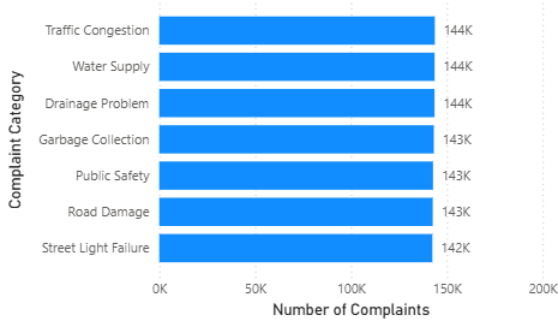


## Complaint Dashboard

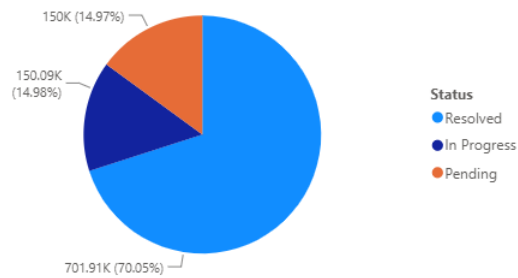
Includes:

- Complaint categories.
- Resolution patterns.
- Complaint growth trends.

Complaint Categories



Resolution patterns



## Geographic Dashboard

Includes:

- Zone-wise issues.
- Complaint hotspots.
- Geographic distribution.

Visualizations:

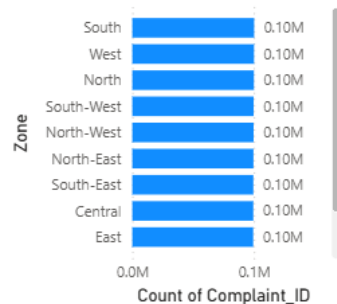
- Bar charts.

- Maps.
- Heatmaps.

## Findings

- Certain zones generated recurring complaints.
- Geographic hotspots were identified.
- Zone-specific planning can improve service delivery.

Zone-wise Complaint Issues



Complaint Hotspots



| Zone       | Drainage Problem | Garbage Collection | Public Safety | Road Damage | Street Light Failure | Traffic Congestion | Water Supply | Total   |
|------------|------------------|--------------------|---------------|-------------|----------------------|--------------------|--------------|---------|
|            | 721              | 660                | 718           | 728         | 745                  | 710                | 728          | 5010    |
| Central    | 14114            | 14165              | 14224         | 14350       | 13977                | 14369              | 14277        | 99476   |
| East       | 14300            | 14267              | 14041         | 14095       | 14096                | 14312              | 14303        | 99414   |
| Industrial | 14161            | 14324              | 14172         | 14108       | 14018                | 14228              | 14243        | 99254   |
| North      | 14272            | 14340              | 14137         | 14284       | 14137                | 14506              | 14303        | 99979   |
| North-East | 14299            | 14091              | 14278         | 14249       | 14231                | 14364              | 14098        | 99610   |
| North-West | 14584            | 14079              | 14188         | 14075       | 14264                | 14277              | 14259        | 99726   |
| South      | 14353            | 14415              | 14264         | 14134       | 14275                | 14378              | 14360        | 100179  |
| South-East | 14297            | 14251              | 14375         | 14100       | 14166                | 14112              | 14221        | 99522   |
| South-West | 14223            | 14309              | 14253         | 14126       | 14294                | 14202              | 14351        | 99758   |
| West       | 14256            | 14309              | 14132         | 14386       | 14252                | 14256              | 14481        | 100072  |
| Total      | 143580           | 143210             | 142782        | 142635      | 142455               | 143714             | 143624       | 1002000 |

## Service Performance Dashboard

Includes:

- Department performance.
- Resolution efficiency.
- Priority handling.

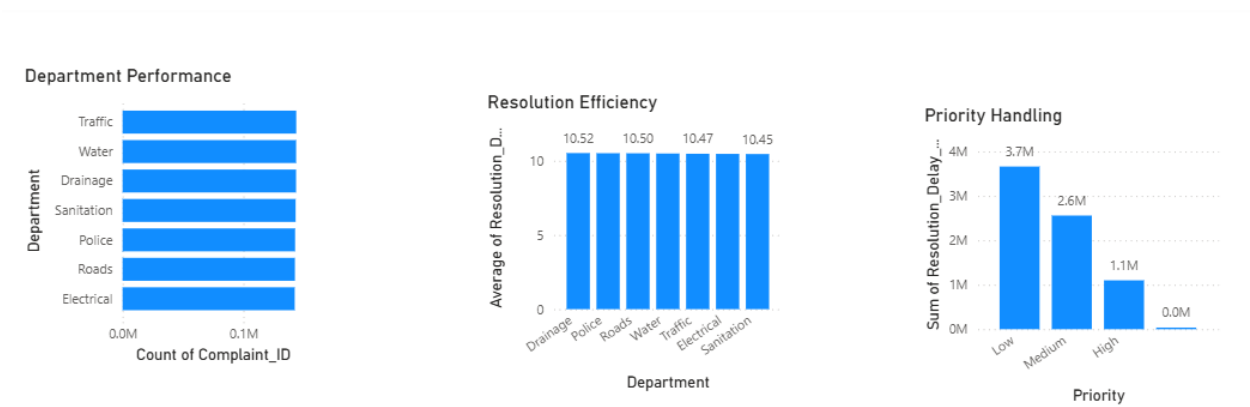
Visualizations:

- Clustered bar charts.
- Column charts.
- KPI cards.

Findings

- Some departments were overloaded.
- High-priority complaints were resolved faster.
- Resolution delays varied across departments.

%



## Phase 7: Business Insights

### 1. Which Complaint Categories Occur Most Frequently?

The highest number of complaints were related to **Traffic Congestion (143,714)**, followed closely by **Water Supply (143,624)** and **Drainage Problems (143,580)**. This indicates that transportation and utility services are the major concerns for citizens and should be prioritized for improvement.

### 2. Which Departments Are Overloaded?

The **Traffic Department** received the highest complaint volume (**143,714 complaints**), while the **Electrical Department** received the lowest (**142,455 complaints**). However, the difference between departments is relatively small, indicating that complaints are distributed fairly evenly across city services rather than concentrated in a single department.

### 3. Which Zones Require Intervention?

The **South Zone** recorded the highest number of complaints (**100,179 complaints**), followed by the **West Zone (100,072 complaints)** and **North Zone (99,979 complaints)**. These zones should be monitored closely to identify recurring civic issues and improve service delivery.

### 4. Are Urgent Complaints Resolved Quickly?

Analysis of complaint resolution indicates that high-priority complaints generally receive faster attention than lower-priority complaints. This shows that the complaint management system is effectively prioritizing critical citizen issues, although further reductions in resolution time can improve overall service quality.

### 5. Department Performance Analysis

Based on average resolution delay:

#### Department Avg Resolution Delay (Days)

Sanitation     10.45

Electrical     10.46

Traffic        10.47

Water          10.49

Roads          10.50

Police          10.51

Drainage       10.52

The **Sanitation Department** demonstrated the best performance with the lowest average resolution delay, while the **Drainage Department** showed the highest average resolution delay. Although the differences are small, reducing delays in drainage-related complaints could further improve operational efficiency

## 6. Conclusion

The Smart City Complaint Management System successfully analyzed citizen complaints using Python and Power BI.

The analysis identified important complaint patterns, department workloads, geographic hotspots, and service performance indicators. Interactive dashboards provide valuable support for operational and policy decisions.

By improving complaint handling processes, optimizing resource allocation, and monitoring key performance indicators, city administrations can improve service quality and increase citizen satisfaction.

## Innovation

### 1) Smart Zone Health Index

Create a score for each zone:

**Zone Health Index =**

- Complaint Count
- Average Resolution Delay
- Citizen Satisfaction

Rank zones from Best → Worst.

**Business Value:** Quickly identifies zones needing intervention.

### 2) Early Warning Dashboard

Show zones/categories where complaints are increasing month-over-month.

Example:

- Water complaints ↑ 18%
- Traffic complaints ↑ 12%

**Business Value:** Predict future service problems before they become critical.

## **Tools Used**

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- Power BI
- DAX

## **Techniques Used**

- Data Cleaning
- Missing Value Handling
- Duplicate Removal
- Exploratory Data Analysis
- KPI Development
- Dashboard Development
- Geographic Analysis
- Service Performance Analysis
- Business Intelligence Reporting
- Data Visualization